

THIAGO J. PINHEIRO DOS SANTOS

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SUMMARY

- I am a statistical mechanics expert and computational chemist who brings physics-based insight into artificial intelligence and machine learning interatomic potentials for simulations of ionic systems, ranging from aqueous paramagnetic ions for medical MRI to molten salts for nuclear energy.

EDUCATION

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- **Ph.D. in Chemical Engineering** Aug/2020 - Apr/2025
Rice University, Houston, TX / USA GPA: 4.0/4.0
Riki Kobayashi Prize Awardee (*Conferred in May/2025)
Advisors: Walter Chapman, Dilip Asthagiri, and Philip Singer
 - **Federal University of Rio de Janeiro**, Rio de Janeiro / Brazil Jan/2018 - Sep/2019
Masters in Chemical and Biochemical Process Engineering 2.88/3.00
Advisors: Frederico Tavares and Charles Abreu
 - **Federal University of Rio de Janeiro**, Rio de Janeiro / Brazil Jan/2014 - Jul/2018
Bachelors in Chemical Engineering 9.2/10.0
Magna cum laude; Distinguished Student Award
 - **Federal Institute of Rio de Janeiro**, Nilopolis / Brazil Jan/2010 - Jul/2014
Associate Degree in Chemistry 9.4/10.0

WORK/RESEARCH EXPERIENCE

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- **Postdoctoral Research Associate** May/2025 – Present
Oak Ridge National Laboratory (ORNL), TN, USA
 - Investigating thermodynamic properties of molten salts used in nuclear reactors by means of quantum mechanics calculations, Machine Learning Force Fields (MLFFs) obtained via *ab initio* data, and molecular dynamics simulations.
 - Pioneering the use of quantum computing to perform high-level quantum calculations of molten salts and accurately describe tritium production in breeder reactors for nuclear fusion.
 - **Graduate Research Assistant** Jan/2021 – May 2025
Rice University, Houston, TX, USA
 - Investigating the NMR relaxation and chemical stability of MRI contrast agents using statistical mechanics, along with quantum and classical modeling. This work enhances the understanding of the underlying physics of NMR relaxation mechanisms and guides the search for newer, safer, and more effective MRI contrast agents.
 - **Graduate Research Assistant** Dec/2017 – Dec/2020
Federal University of Rio de Janeiro, RJ, Brazil
 - Investigating the mass transport properties of hydrocarbon-CO₂ systems at high pressure via molecular simulations, as well as how polyethoxylated surfactants can affect interfacial properties of water/oil and oil/water emulsions.
 - **Research Intern** Dec/2017 – Jul/2018
UFRJ/PETROBRAS (CENPES), RJ, Brazil

- Undergraduate internship on a joint project at the CENPES research center of PETROBRAS (the 70th-largest public company in the world) and UFRJ. The work consisted of computational research on the effect of thermodiffusion on paraffin deposition, leading to a better description of the thermodynamic models used to predict paraffin deposition in oil pipelines.

PUBLICATIONS

1.  S. Das, **T. J. Pinheiro dos Santos**, S. Bhowmik, M. Bazayeva, Z. Li, A. Shajan, D. Kaliakin, F. Liang, V. S. Bryantsev, A. Geist, A. McClain Gomez, T. Pellegrini, R. Walkup, S. R. Seelam, M. Motta, K. M. Merz Jr., and T. L. Beck. Quantum Computations on Fusion Blanket Molten Salts. *arXiv:2606.30402* (2026).
2.  **T. J. Pinheiro dos Santos**, L. D. Gibson, R. Chahal-Crockett, S. Banerjee, M. Taillefumier, A. Sedova, J. Schorne-Pinto, R. Messerly, and V. S. Bryantsev. Stability and thermophysical properties of solid and molten BeF₂ via density functional theory and machine learning potentials. *Digital Discovery* (2026).
3.  K. Buttice, S. Banerjee, **T. J. Pinheiro dos Santos**, E. Kim, Q. Yang, A. Ramos-Ballesteros, S. Dai, R. Gakhar, S. Roy, V. S. Bryantsev, and A. Couet. In Situ UV–vis Quantification of Chlorobasicity in Binary Molten Chlorides. *The Journal of Physical Chemistry B* (2026).
4.  D. Asthagiri, **T. J. Pinheiro dos Santos**, and T. L. Beck. Equipartition and the Temperature of Maximum Density of TIP4P/2005 Water. *The Journal of Physical Chemistry B*, 130, 862–867 (2026).
5.  **T. J. Pinheiro dos Santos**, B. Orcan-Ekmekci, W. G. Chapman, P. M. Singer, and D. Asthagiri. Extended molecular eigenmodes treatment of dipole-dipole NMR relaxation in real fluids. *The Journal of Chemical Physics*, 163, 184105 (2025).
6.  **T. J. Pinheiro dos Santos**, C. C. Fraenza, G. A. L. Souza, E. Pelegano-Titmuss, D. Asthagiri, S. G. Greenbaum, W. G. Chapman, and P. M. Singer. Molecular-Level Insights into the NMR Relaxivity of Gadobutrol using Quantum and Classical Molecular Simulations. *Chemical & Biomedical Imaging*, 3(9), 615–629 (2025).
7.  A. Valiya Parambathu, **T. J. Pinheiro dos Santos**, W. G. Chapman, G. J. Hirasaki, D. Asthagiri, and P. M. Singer. Molecular Modes Elucidate the Nuclear Magnetic Resonance Relaxation of Viscous Fluids. *The Journal of Physical Chemistry B*, 128, 8017–8028 (2024).
8.  **T. J. Pinheiro dos Santos**, B. Orcan-Ekmekci, W. G. Chapman, P. M. Singer, and D. Asthagiri. Theory and Modeling of Molecular Modes in the NMR Relaxation of Fluids. *The Journal of Chemical Physics*, 160, 064108 (2024).
9.  **T. J. Pinheiro dos Santos**, A. Valiya Parambathu, C. C. Fraenza, C. Walsh, S. G. Greenbaum, W. G. Chapman, D. Asthagiri, and P. M. Singer. Thermal and Concentration Effects on ¹H NMR Relaxation of Gd³⁺-aqua using MD Simulations and Measurements. *Physical Chemistry Chemical Physics*, 24, 27964–27975 (2022).
10.  A. M. Luz, **T. J. Pinheiro dos Santos**, G. D. Barbosa, C. L. M. Camargo, and F. W. Tavares. A Molecular Study on the Behavior of Polyethoxylated Alkyl Ethers Surfactants in a Water/n-alkane Interface. *Colloids and Surfaces A*, 650, 129627 (2022).
11.  **T. J. Pinheiro dos Santos** (co-first author), A. Valiya Parambathu, W. G. Chapman, and D. Asthagiri. Comment on “Calculation of Solid–Fluid Interfacial Free Energy with Consideration of Solid Deformation by Molecular Dynamics”. *The Journal of Physical Chemistry A*, 126, 1782–1783 (2022).
12.  P. M. Singer, A. Valiya Parambathu, **T. J. Pinheiro dos Santos**, Y. Liu, L. B. Alemany, G. J. Hirasaki, W. G. Chapman, and D. Asthagiri. Predicting ¹H NMR Relaxation in Gd³⁺-aqua using Molecular Dynamics Simulations. *Physical Chemistry Chemical Physics*, 23, 20974–20984 (2021).

13.  **T. J. Pinheiro dos Santos**, C. R. A. Abreu, and F. W. Tavares. Fick Diffusion Coefficients via Molecular Dynamics: An Alternative Approach in the Fourier Domain. *Journal of Molecular Liquids*, 329, 115460 (2021).
14.  **T. J. Pinheiro dos Santos**, C. R. A. Abreu, B. A. C. Horta, and F. W. Tavares. Self-diffusion Coefficients of Methane/n-hexane Mixtures at High Pressures: An Evaluation of the Finite-size Effect and a Comparison of Force Fields. *The Journal of Supercritical Fluids*, 155, 104639 (2020).

HONORS AND AWARDS

- **Best Oral Presentation Award - Nuclear Materials Session**
14th Annual ORPA Research Symposium, TN/USA May/2026
- **Best Postdoctoral Poster Award**
Molten Salt Reactor Workshop 2025, TN/USA Nov/2025
- **Finalist for the university-wide Graduate Teaching Award**
Center for Teaching Excellence, Rice University, TX/USA Mar/2024
- Annually granted to the best Graduate Teaching Assistant across all fields at Rice University.
- **Diploma *Magna cum laude***
Federal University of Rio de Janeiro, Brazil Dec/2019
- **Distinguished Student Award**
School of Chemistry, Federal University of Rio de Janeiro, Brazil Dec/2018
- Annually granted to a few top students with outstanding academic performance among the all students in the School of Chemistry.
- **Best Research Work at the XXXVIII Giulio Massarani Scientific Initiation Fair**
Center of Technology, Federal University of Rio de Janeiro, Brazil Oct/2016
- This award is annually granted to the best research work among the all engineering students in the Center of Technology.

ACTIVITIES AND LEADERSHIP

- Discussion Leader for the “Interdisciplinary Innovations with Ionic Liquids and Molten Salts”
Gordon Research Symposium (GRS) on Ionic Liquids, ME/USA Aug/2026
- Organizer/student leader for the activities of the Graduate International Orientation Week
Rice University, TX/USA Jul/2022 – Jul/2024
- Secretary and Treasurer of the Brazilian Student Association (BRASA) at Rice University
Rice University, TX/USA Aug/2022 – Aug/2024
- Secretary in the Chemical and Biomolecular Engineering Graduate Student Association
Rice University, TX/USA Jul/2021 – Jul/2023
- Member of the Organizing Committee in the X Brazilian Conference on Applied Thermodynamics
CBTermo, Nova Friburgo, Brazil Nov/2019

INVITED TALKS

1. **T. J. Pinheiro dos Santos**. Towards molecular understanding of MRI contrast agents and NMR relaxivity using classical and quantum modeling. In: ATOMS Seminar Series. Federal University of Rio de Janeiro, Brazil (2025).
2. **T. J. Pinheiro dos Santos**. Towards design of MRI contrast agents to improve NMR relaxivity using classical and quantum modeling. In: Statistical Thermodynamics & Molecular Simulations (STMS) Seminar Series. Yale University and University of Minnesota - Twin Cities, USA (2024).

SKILLS

- **Languages:** Portuguese (native speaker), Spanish (fluent), English (fluent), and German (basic).
- **Programming Languages:** Python, FORTRAN, MATLAB, C, and BASH.
- **Molecular Modeling packages:** LAMMPS, OpenMM, VASP, ASE (Molecular Dynamics), Cassandra (Monte Carlo), Gaussian, PSI4, ORCA, CP2K (Quantum simulations), VMD, Playmol/Packmol, Cluster Management Packages.

FELLOWSHIPS

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| • Riki Kobayashi Fellowship Prize in Chemical Engineering | Dec/2023 - Apr/2025 |
| • Future Faculty Fellowship at Rice University | Jul/2024 - Apr/2025 |
| • Ken Kennedy Institute Fellowship | Jan/2022 |
| • PETROBRAS Graduate Researcher Fellowship | Oct/2018 – Dec/2020 |
| • CAPES Graduate Researcher Fellowship | Sep/2018 – Sep/2019 |

TRAVEL GRANTS AND SCHOLARSHIPS

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| • Rice Engineering Alumni (REA) Travel Grant | Nov/2023 |
| • i-CoMSE/OSU Travel Grant for MC/MD Summer School | Jul/2022 |
| • COPPETEC Travel Grant for International Scientific Conferences | May/2019 |
| • COPPETEC Travel Grant for International Scientific Conferences | Oct/2018 |
| • CNPq Scientific Initiation Scholarships | Mar/2016 – Aug/2018 |
| • CAPES Young Talents for Science Scholarships | Mar/2015 – Feb/2016 |
| • IFRJ Student Scholarships | Jan/2014 – Dec/2014 |

TEACHING EXPERIENCE

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| • Teaching Assistant for undergraduate-level Thermodynamics (CHBE 412), undergraduate level Applied Mathematics and Numerical Methods (CHBE 305), graduate-level Advanced Thermodynamics (CHBE 611), graduate-level Kinetics, Catalysis, and Reactor Engineering (CHBE 590), and graduate-level Advanced Numerical Methods (CHBE 505)
Rice University, TX/USA | Jan/2021 – May/2023 |
| • Instructor of the Molecular Simulation Course in the Online Pilot School Prof. Giulio Massarani
Federal University of Rio de Janeiro, Brazil | Jul/2019 – Jul/2020 |
| • Co-instructor for undergraduate level Statistical Thermodynamics
Federal University of Rio de Janeiro, Brazil | Jan/2018 – Sep/2019 |